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Electronic device receiving wireless power

Abstract

A wireless power receiver receiving power from a wireless power transmitter is provided. The receiver includes a resonance circuit, a rectifier circuit, and a driver circuit. The resonance circuit includes first and second coils and a first capacitor. The rectifier circuit includes first and second rectifier circuits. The first rectifier circuit includes first through fourth MOSFETs. Sources of the first and second MOSFETs are connected to ends of a resonator including the first coil and the first capacitor. Sources of the third and fourth MOSFETs are connected to ground. The driver circuit is connected to gates of the first through fourth MOSFETs. When the driver circuit switches off the first and second MOSFETs and switches on the third and fourth MOSFETs, as currents are induced in the resonator and the second coil, the resonance circuit receives the wireless power, and the current induced in the second coil is rectified by the second rectifier circuit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a Continuation of application Ser. No. 17/523,476, filed Nov. 10, 2021 (now U.S. Pat. No. 11,996,704), which is a continuation of International Application No. PCT/KR2021/014795, which was filed on Oct. 21, 2021, and is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2020-0154941, which was filed in the Korean Intellectual Property Office on Nov. 18, 2020, and Korean Patent Application No. 10-2021-0065693, which was filed in the Korean Intellectual Property Office on May 21, 2021, the entire disclosure of each of which is incorporated herein by reference.

BACKGROUND

1. Field

(1) The disclosure relates generally to an electronic device receiving wireless power, and more particularly, to a wireless power receiver having a bidirectional switch.

2. Description of Related Art

(2) Wireless charging technology that uses an electromagnetic induction scheme or a magnetic resonance scheme has been utilized on electronic devices such as smart phones. When a power transmitting unit (PTU) (e.g., a wireless charging pad) and a power receiving unit (PRU) (e.g., a smart phone) come into contact with each other or approach each other within a certain distance, a battery of the PRU may be charged by electromagnetic induction or electromagnetic resonance between a transmitting coil of the PTU and a receiving coil of the PRU.

(3) FIG. 1 is a circuit diagram illustrating a wireless power receiver performing wireless charging. A wireless power receiver **100** may include a resonance circuit **110**, a first rectifier circuit **130**, a second rectifier circuit **140**, a capacitor **151**, a low drop-output (LDO) regulator **152**, a charger **153**, a battery **154**, a switching circuit **170**, and a controller **180**. The resonance circuit **110** may include a first coil **111**, a first capacitor **112**, a second coil **113**, and a bidirectional switch **120**. The second coil **113** may be a coil having an inductance that is smaller than that of the first coil **111**. The first coil **111** may be a receiver (RX) resonant coil forming parallel resonance with the first capacitor **112**, and the second coil **113** may be an RX auxiliary coil or feeding coil connected to the second rectifier circuit **140**. The first coil **111** and the second coil **113** may be magnetically coupled to each

other, such that a mutual inductance $M_{sub.f}$ may be formed between the first coil **111** and the second coil **113**. Due to the formed mutual inductance $M_{sub.f}$, a voltage may be induced in the second coil **113** that is a voltage induced in the first coil **111** increased by a goodness Q of the resonance circuit. A current may then flow through the second rectifier circuit **140** due to the voltage induced in the second coil **113**.

(4) As will be described in greater detail below with reference to FIGS. 2A and 2B, a connection state between the first coil **111**, the first capacitor **112**, and the second coil **113**, constituting the resonance circuit **110**, may differ with an on/off state of the bidirectional switch **120**, and wireless power received by the resonance circuit **110** from the wireless power transmitter and transmitted to the second rectifier circuit **140** and the first rectifier circuit **130** may differ with the connection state between the first coil **111**, the first capacitor **112**, and the second coil **113**.

(5) The bidirectional switch **120** may include two or more switches (e.g., a first switch **121** and a second switch **122**). Each of the first switch **121** and the second switch **122** may be an N-channel metal-oxide semiconductor field-effect-transistor (MOSFET), and a first diode **123** and a second diode **124** may be a body diode of the first switch **121** and a body diode of the second switch **122**, respectively. The first switch **121** and the second switch **122** may have a common source voltage by having respective resources serially connected at a node **127**. A common gate voltage with the common source voltage as a reference voltage may be applied to a gate of each of the first switch **121** and the second switch **122**, under control of a gate driver circuit **160**. For example, an induced voltage may be formed in the first coil **111** based on a high resonant frequency (e.g., 6.78 MHz), such that source voltages of the first switch **121** and the second switch **122** may be alternated depending on a high frequency (e.g., 6.78 MHz). By applying the gate voltage with the common source voltage as the reference voltage to the gate of each of the first switch **121** and the second switch **122**, the on/off states of the first switch **121** and the second switch **122** may be stably controlled. Applying the gate voltage or applying (or outputting) a driving signal may mean applying a voltage exceeding each threshold voltage to the gates of the first switch **121** and the second switch **122**. Not applying the gate voltage or not applying the driving signal may mean applying a voltage that does not exceed each threshold voltage to the gates of the first switch **121** and the second switch **122**. When the gate voltage is applied to the gate of each of the first switch **121** and the second switch **122**, the first switch **121** and the second switch **122** may be switched to an “on” state and both ends of the bidirectional switch **120** may be electrically connected. When the gate voltage is not applied to the gate of each of the first switch **121** and the second switch **122**, the first switch **121** and the second switch **122** may be switched to an “off” state and an electrical connection path between both ends of the bidirectional switch **120** may be cut off.

(6) The first rectifier circuit **130** may include a third diode **131**, a fourth diode **132**, and/or a capacitor **133**. When the first and switch **121** and the second switch **122** are switched from the “on” state to the “off” state, upon application of a voltage of a predetermined magnitude or greater to the both ends of the first switch **121** and the second switch **122**, the third diode **131** and the fourth diode **132** may operate to rectify an induced voltage, and thus, apply the rectified voltage as a both-end voltage of the capacitor **133**. The capacitor **133** may be charged with the voltage rectified by the third diode **131** and the fourth diode **132**. The capacitor **133** may be connected to the node **127** of the first switch **121** and the second switch **122**, and may be applied at one end thereof with the source voltages of the first switch **121** and the second switch **122**.

(7) After the capacitor **133** is charged with the voltage rectified by the third diode **131** and the fourth diode **132** or a voltage spike generated in both ends of the first switch **121** and the second switch **122**, charged energy may be discharged through at least one resistor connected in parallel to the capacitor **133**.

(8) The second rectifier circuit **140** may include four diodes **141**, **142**, **143**, and **144**, and rectify the induced voltage to apply the rectified voltage as a both-end voltage of the capacitor **151**.

(9) The first rectifier circuit **130** may form an energy harvesting circuit that charges the capacitor

133. The energy harvesting circuit may include the first rectifier circuit **130**, a Zener diode D.sub.z1 **101** connected in parallel to the capacitor **133**, and a Zener diode D.sub.z2 **103**. One end of each of the capacitor **133**, the Zener diode D.sub.z1 **101**, and the Zener diode D.sub.z2 **103** may be connected to the node **127** that is the common source of the first switch **121** and the second switch **122**.

(10) The Zener diode **101** may regulate the rectified voltage applied to the capacitor **133** to a predetermined voltage (e.g., a Zener breakdown voltage of the Zener diode **101**) or less. A resistor R.sub.g **102** may be a gate resistor of the first switch **121** and the second switch **122**, and a capacitor C.sub.gs **104** may be a gate-source capacitor of the first switch **121** and the second switch **122**. The energy harvesting circuit may further include the Zener diode D.sub.z2 **103** connected in parallel to a gate and a source of the first switch **121** and/or a gate and a source of the second switch **122**. The Zener diode **103** may regulate the voltage applied to the capacitor **104** to a predetermined voltage (e.g., a Zener breakdown voltage of the Zener diode **103**) or less. The switching circuit **170** may include an optocoupler Q.sub.C1 **175** and a switch Q.sub.C2 **176**. A diode **177** may be a body diode of the switch **176**. The optocoupler **175** may include a light-emitting diode (LED) **171** and a photodiode **173**. A resistor R.sub.P **174** may be an internal resistor of the photodiode **173**. A resistor R.sub.C **172** may be an internal resistor of the LED **171** of the optocoupler **175**.

(11) The switching circuit **170** may be provided with a first control signal from the controller **180** in the form of a voltage V.sub.con applied to the switch **176** of the optocoupler **175**. The controller **180** may be supplied with power for applying the voltage V.sub.con from the battery **154** at all times. The controller **180** may be supplied with power for applying the voltage V.sub.con based on the LDO regulator **152** when receiving wireless power.

(12) The controller **180** may compare a rectified voltage V.sub.rec of an output end of the second rectifier circuit **140** with a threshold, and apply the voltage V.sub.con to the switch **176** based on a comparison result. The controller **180** may apply the voltage V.sub.con exceeding a threshold voltage of the switch **176** when determining that the rectified voltage V.sub.rec at the output end of the second rectifier circuit **140** is greater than or equal to the threshold. The optocoupler **175** may operate based on the voltage V.sub.con applied from the controller **180** to the switch **176**. When a magnitude of the voltage V.sub.con applied from the controller **180** to the switch **176** exceeds the threshold voltage of the switch **176**, the switch **176** may be switched to the “on” state, such that the LED **171** may be connected to the ground through the switch **176**. As the LED **171** is connected to the ground through the switch **176**, the LED **171** may be switched to the “on” state based on a voltage of a node **178** to radiate light. The voltage of the node **178** may be, for example, the rectified voltage V.sub.rec at the output end of the second rectifier circuit **140**. Alternatively, a voltage from an external power source other than the rectified voltage V.sub.rec at the output end of the second rectifier circuit **140** may be applied to the LED **171**. The photodiode **173** may be switched to the “on” state when the light radiated from the LED **171** arrives. As the photodiode **173** is switched to the on state, a connection path may be formed between the resistor **174** and the sources of the first switch **121** and the second switch **122**. As a current flows along the connection path between the resistor **174** and the sources of the first switch **121** and the second switch **122**, the charge of the capacitor **133** may be discharged. Thus, when the both-end voltage of the capacitor **104**, which is equal to the voltage between the gates and the sources of the first switch **121** and the second switch **122**, gradually decreases to below threshold voltages of the first switch **121** and the second switch **122**, the first switch **121** and the second switch **122** may be switched to the “off” state.

(13) The controller **180** may control the voltage V.sub.con applied to the switch **176** to be substantially 0, when determining that the rectified voltage V.sub.rec at the output end of the second rectifier circuit **140** is less than the threshold. When the magnitude of the voltage V.sub.con applied from the controller **180** to the switch **176** does not exceed the threshold voltage of the

switch **176**, the switch **176** may be switched to the off state, such that the LED **171** may be switched to the “off” state and light may not be radiated from the LED **171**. The photodiode **173** may be switched to the “off” state when the light radiated from the LED **171** does not arrive. As the photodiode **173** is switched to the “off” state, the connection between the resistor **174** and the sources of the first switch **121** and the second switch **122** may be cut off, and the capacitor **104** may be charged by energy stored in the capacitor **133**. When the both-end voltage of the capacitor **104** increases above the threshold voltages of the first switch **121** and the second switch **122**, the first switch **121** and the second switch **122** may be switched to the on state.

(14) When the magnitude of the voltage $V_{sub.con}$ applied from the controller **180** to the switch **176** of the switching circuit **170** exceeds the threshold voltage of the switch **176**, the both-end voltage of the capacitor **104** may decrease below the threshold voltages of the first switch **121** and the second switch **122**, such that the first switch **121** and the second switch **122** may be switched to the “off” state. On the other hand, when the magnitude of the voltage $V_{sub.con}$ applied from the controller **180** to the switch **176** of the switching circuit **170** is less than the threshold voltage of the switch **176**, the both-end voltage of the capacitor **104** may increase above the threshold voltages of the first switch **121** and the second switch **122**, such that the first switch **121** and the second switch **122** may be switched to the “on” state. That is, the switching circuit **170** may output a control signal for controlling the bidirectional switch **120** including the first switch **121** and the second switch **122** in the form of the both-end voltage of the capacitor **104**, according to the voltage $V_{sub.con}$ applied to the switch **176** from the controller **180**.

(15) FIG. 2A is a diagram illustrating an equivalent circuit of the wireless power receiver when the magnitude of the voltage $V_{sub.con}$ applied from the controller exceeds the threshold voltage of the switch. FIG. 2B is a diagram illustrating an equivalent circuit of the wireless power receiver when the magnitude of the voltage $V_{sub.con}$ applied from the controller is less than the threshold voltage of the switch. In equivalent circuit diagrams of FIGS. 2A and 2B, a receiving circuit **210** may include the second rectifier circuit **140**, the capacitor **151**, the LDO regulator **152**, the charger **153**, and the battery **154** of FIG. 1.

(16) Referring to FIG. 2A, as described with reference to FIG. 1, when the magnitude of the voltage $V_{sub.con}$ applied from the controller **180** to the switch **176** of the switching circuit **170** exceeds the threshold voltage of the switch **176**, the bidirectional switch **120** including the first switch **121** and the second switch **122** may be switched to the “off” state such that a current may not flow through both ends of the bidirectional switch **120**. In this case, the receiving circuit **210** may be serially connected to the first coil **111**, the first capacitor **112**, and the second coil **113** to form a serial resonance circuit, and the receiving circuit **210**, the first coil **111**, the first capacitor **112**, and the second coil **113** may form one closed loop **200a**. In addition, based on a magnetic field generated from a power transmitter, a current may be induced in a circuit including the first coil **111**, the first capacitor **112**, and the second coil **113** of the resonance circuit **110**, such that the resonance circuit **110** may receive wireless power.

(17) Referring to FIG. 2B, as described with reference to FIG. 1, when the magnitude of the voltage $V_{sub.con}$ applied from the controller **180** to the switch **176** of the switching circuit **170** is less the threshold voltage of the switch **176**, the bidirectional switch **120** including the first switch **121** and the second switch **122** may be switched to the “on” state such that a current may flow through both ends of the bidirectional switch **120**. In this case, the receiving circuit **210** may be connected in parallel to a first circuit, which is formed by serially connecting the first coil **111** and the first capacitor **112**, and the second coil **113** to form a parallel resonance circuit. The first coil **111** and the first capacitor **112** may form one closed loop **201b**, and the receiving circuit **210** and the second coil **113** may form one closed loop **202b**. In addition, a current may be induced in the first circuit including the first coil **111** and the first capacitor **112** of the resonance circuit **110** based on a magnetic field generated from a power transmitter, and a current may be induced to a second circuit including the second coil **113** based on a magnetic field generated in the first circuit, such

that the resonance circuit **110** may receive wireless power.

(18) The wireless power receiver **100**, described with reference to FIGS. **1**, **2A**, and **2B**, requires the gate driver circuit **160** for controlling the gate voltages of the first switch **121** and the second switch **122**, which limits downsizing of the wireless power receiver **100**. Moreover, as the gate driver circuit **160** includes many elements, the wireless power receiver **100** is also limited in any reduction in production cost.

SUMMARY

(19) In accordance with an aspect of the disclosure, a wireless power receiving device may include a bidirectional switch, a source of which may be connected to the ground.

(20) In accordance with an aspect of the disclosure, a wireless power receiver is provided that receives wireless power from a wireless power transmitter. The wireless power receiver includes a resonance circuit configured to receive the wireless power, and including a first coil, a second coil, and a first capacitor. The wireless power receiver also includes a rectifier circuit connected to the resonance circuit, and including a first rectifier circuit and a second rectifier circuit forming a full bridge structure. The wireless power receiver also includes a driver circuit. The first rectifier circuit includes a first MOSFET, a second MOSFET, a third MOSFET, and a fourth MOSFET. Sources of the first MOSFET and the second MOSFET are connected to both ends of a resonator in which the first coil and the first capacitor are connected in series. Sources of the third MOSFET and the fourth MOSFET are connected to ground. The driver circuit is connected to gates of the first MOSFET, the second MOSFET, the third MOSFET and the fourth MOSFET. When the driver circuit switches off the first MOSFET and the second MOSFET and switches on the third MOSFET and the fourth MOSFET, as a first current is induced in the resonator based on a first magnetic field generated from the wireless power transmitter and a second current is induced in the second coil based on a second magnetic field generated in the resonator, the resonance circuit receives the wireless power, and the second current induced in the second coil is rectified by the second rectifier circuit.

(21) In accordance with an aspect of the disclosure, a wireless power receiver is provided that receives wireless power from a wireless power transmitter. The wireless power receiver includes a resonance circuit configured to receive the wireless power, and includes a first coil, a second coil, a first capacitor, and bidirectional switch. The wireless power receiver also includes a rectifier circuit connected to the resonance circuit, and a driver circuit. The bidirectional switch includes a first MOSFET and a second MOSFET. Sources of the first MOSFET and the second MOSFET are connected to ground. The driver circuit is connected to gates of the first MOSFET and the second MOSFET. When the driver circuit switches off the first MOSFET and the second MOSFET, as a first current is induced in a first circuit including the first coil, the second coil, and the first capacitor based on a first magnetic field generated from the wireless power transmitter, the resonance circuit is configured to receive the wireless power. When the driver circuit switches on the first MOSFET and the second MOSFET, as a second current is induced in a second circuit including the first coil and the first capacitor based on the first magnetic field generated from the wireless power transmitter and a third current is induced in a third circuit including the second coil based on a second magnetic field generated in the second circuit, the resonance circuit is configured to receive the wireless power.

(22) According to embodiments of the disclosure, a wireless power receiver is provided that receives wireless power from a wireless power transmitter. The wireless power receiver includes a bidirectional switch, a source of which may be connected to the ground. The wireless power receiver may not need the complex gate driver circuit **160** of FIG. **1**, and thus, may be produced at a low unit cost with a small size.

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a circuit diagram illustrating a wireless power receiver that performs wireless charging;

(3) FIGS. 2A and 2B are diagrams illustrating circuits of a wireless power receiver;

(4) FIG. 3 is a circuit diagram illustrating a wireless power transmitter and a wireless power receiver, according to an embodiment;

(5) FIG. 4 is a circuit diagram illustrating a wireless power receiver, according to an embodiment;

(6) FIGS. 5A and 5B is a diagram illustrating equivalent circuits of a wireless power receiver, according to an embodiment;

(7) FIG. 6 is a circuit diagram illustrating a wireless power receiver, according to an embodiment;

(8) FIGS. 7A and 7B is a diagram illustrating equivalent circuits of a wireless power receiver, according to an embodiment;

(9) FIG. 8 is a diagram illustrating components for controlling a driver circuit of a wireless power receiver, according to an embodiment;

(10) FIG. 9 is a circuit diagram illustrating a wireless power receiver, according to an embodiment;

(11) FIGS. 10A and 10B are diagrams illustrating equivalent circuits of a wireless power receiver, according to an embodiment;

(12) FIG. 11 is a circuit diagram illustrating a wireless power receiver, according to an embodiment;

(13) FIG. 12 is a circuit diagram illustrating a wireless power receiver, according to an embodiment; and

(14) FIGS. 13A and 13B are diagrams illustrating equivalent circuits of a wireless power receiver, according to an embodiment.

DETAILED DESCRIPTION

(15) The disclosure may have various embodiments, and various modifications and changes may be made to the embodiments of the disclosure. Hereinafter, some of the embodiments will be described in detail with reference to the accompanying drawings. However, this is not intended to limit the disclosure to particular embodiments, and it should be appreciated that the disclosure includes all changes, equivalents, or alternatives falling within the technical idea and scope of the disclosure.

(16) With regard to the description of the drawings, similar reference numerals may be used to designate similar or relevant elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. The terms including an ordinal number, such as expressions “a first” and “a second” may be used to describe various elements, but the corresponding elements should not be limited by such terms. These terms are used merely to distinguish between one element and any other element. For example, a first element may be termed a second element, and similarly, a second element may be termed a first element without departing from the scope of the disclosure. The term “and/or” includes any combination of a plurality of relevant items or nay one of the plurality of relevant items. When an element (e.g., first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., second element), it means that the element may be coupled or connected to or with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(17) Further, relative terms described as viewed from the drawings, such as “front surface”, “rear surface”, “top surface”, and “bottom surface”, may be replaced by ordinal numbers such as “first” and “second”. In the ordinal numbers such as “first” and “second”, the order of them is determined in the mentioned order or arbitrarily and may be arbitrarily changed as necessary.

(18) The terms used in the disclosure are only used to describe specific embodiments, and are not intended to limit the disclosure. A singular expression may include a plural expression unless they are definitely different in a context. In the disclosure, it should be appreciated that the expression “include” or “have” is intended to refer to the existence of a corresponding feature, numeral, step, operation, constituent element, component, or a combination thereof, and does not exclude the existence or addition of one or more other features, numerals, steps, operations, constituent elements, components, or a combination thereof.

(19) Unless defined otherwise, all terms used herein, including technical and scientific terms, have the same meaning as those commonly understood by a person skilled in the art to which the disclosure pertains. Such terms as those defined in a generally used dictionary should be interpreted to have the meanings equal to the contextual meanings in the relevant field of art, and are not to be interpreted to have ideal or excessively formal meanings unless clearly defined in the disclosure.

(20) In the disclosure, the electronic device may be any device including a touch panel, and the electronic device may be referred to as a terminal, a portable terminal, a mobile terminal, a communication terminal, a portable communication terminal, a portable mobile terminal, or a display device.

(21) For example, the electronic device may include a smartphone, a mobile phone, a navigation device, a game console, a TV, an in-vehicle head unit, a notebook computer, a laptop computer, a tablet computer, a personal media player (PMP), or a personal digital assistant (PDA). The electronic device may be implemented as a pocket-sized portable communication terminal having a wireless communication function. In addition, the electronic device may be a flexible device or a flexible display device.

(22) The electronic device may communicate with an external electronic device, such as a server, or may perform a task through interworking with an external electronic device. For example, the electronic device may transmit an image captured using a camera and/or location information detected by a sensor unit to a server through a network. The network may be, but is not limited to, a mobile or cellular network, a local area network (LAN), a wireless local area network (WLAN), a wide area network (WAN), the Internet, or a small area network (SAN).

(23) According to embodiments of the disclosure, a wireless power receiver is provided that receives wireless power from a wireless power transmitter. The wireless power receiver includes a bidirectional switch, a source of which may be connected to the ground. The wireless power receiver may not need the complex gate driver circuit **160** of FIG. **1**, and thus, may be produced at a low unit cost with a small size.

(24) FIG. **3** is a circuit diagram illustrating a wireless power transmitter and a wireless power receiver, according to an embodiment of the disclosure.

(25) Referring to FIG. **3**, a wireless power transmitter **301** may include a power source **311**, an inverter **312**, and a resonance circuit. The resonance circuit of the wireless power transmitter **301** may include a first capacitor **313**, a second capacitor **314**, a resistor **315**, and a transmitting coil **316**. The power source **311** may output direct current (DC) power, and the inverter **312** may convert the DC power output from the power source **311** into alternating current (AC) power. Characteristic values of the first capacitor **313**, the second capacitor **314**, the resistor **315**, and the transmitting coil **316** constituting the resonance circuit may be set in consideration of a resonance frequency and/or impedance matching. At least one of the first capacitor **313**, the second capacitor **314**, the resistor **315**, and the transmitting coil **316** constituting the resonance circuit may be implemented as a variable element having a characteristic value that may be changed. As shown in FIG. **3**, it may be understood by those of ordinary skill in the art that the connection between the

first capacitor **313**, the second capacitor **314**, the resistor **315**, and the transmitting coil **316** is merely exemplary.

(26) Referring to FIG. 3, according to various embodiments, the wireless power receiver **302** may include a resonance circuit, a rectifier circuit **340**, a capacitor **351**, and a load **352**. The resonance circuit may include a first receiving coil **321**, a second receiving coil **322**, a first switch **323**, a second switch **324**, a first resistor **325**, a second resistor **326**, a first capacitor **331**, a second capacitor **332**, and a third capacitor **333**. The first switch **323** and the second switch **324** may be implemented as MOSFETs.

(27) According to various embodiments, sources of the first switch **323** and the second switch **324** may be connected to the ground. Thus, the wireless power receiver **302** may not need an isolated gate driver circuit (e.g., the gate driver circuit **160** of FIG. 1) to drive the first switch **323** and the second switch **324**.

(28) According to various embodiments, the first receiving coil **321** and the second receiving coil **322** of the resonance circuit may be magnetically coupled to the transmitting coil **316** of the wireless power transmitter **301**. When the first switch **323** and the second switch **324** are switched to the “on” state, an induced current based on a magnetic field induced in the transmitting coil **316** may flow through the first receiving coil **321** and the second receiving coil **322** of the resonance circuit, which may be referred to as the resonance circuit receiving the wireless power from the wireless power transmitter **301**. The wireless power received in the resonance circuit may be rectified into DC by the rectifier circuit **340**. The rectifier circuit **340** may include four diodes **341**, **342**, **343**, and **344** forming a full bridge structure. The rectifier circuit **340** may instead include four MOSFETs forming a full bridge structure. The rectifier circuit **340** may also instead include two FETs or two diodes forming a half bridge structure.

(29) According to various embodiments, based on the wireless power rectified in the form of DC by the rectifier circuit **340**, the capacitor **351** may be charged and power may be supplied to the load **352**. The load **352** may include a DC/DC converter, an LDO regulator, a charger, a battery, and a driver circuit that generates a driving signal for switching the first switch **323** and the second switch **324**.

(30) According to various embodiments, when the first switch **323** and the second switch **324** are switched to the “off” state, a current may not flow through the first receiving coil **321** and the second receiving coil **322** of the resonance circuit. At this time, power may be supplied to the load **352** based on charges charged in the capacitor **351**.

(31) According to various embodiments, a control circuit may operate to prevent an excessively high voltage from being applied to the wireless power receiver **302**. For example, the control circuit may switch the first switch **323** and the second switch **324** to the “off” state when a voltage applied to an output end of the rectifier circuit **340** is higher than a threshold. The control circuit may switch the first switch **323** and the second switch **324** to the “on” state when the voltage applied to the output end of the rectifier circuit **340** is less than or equal to the threshold.

(32) FIG. 4 is a circuit diagram illustrating a wireless power receiver, according to an embodiment of the disclosure. FIGS. 5A and 5B are diagrams illustrating equivalent circuits of a wireless power receiver, according to an embodiment of the disclosure.

(33) Referring to FIG. 4, a wireless power receiver **401** may include a resonance circuit, a rectifier circuit **420**, a driver circuit **419**, and a capacitor **430**. The wireless power receiver **401** may further include a load that includes a DC/DC converter, an LDO regulator, a charger, and a battery.

(34) According to various embodiments, the resonance circuit may include a first coil **411**, a first capacitor **412**, a second coil **413**, a third coil **414**, a second capacitor **417**, and a third capacitor **418**, and a bidirectional switch. According to various embodiments, the bidirectional switch may be implemented with a first MOSFET **415** and a second MOSFET **416**. The second coil **413** and the third coil **414** may be wound such that a dot direction is indicated opposite to a direction shown in FIG. 4.

(35) According to various embodiments, sources of the first MOSFET **415** and the second MOSFET **416** may be connected to the ground. Thus, the wireless power receiver **401** may not need an isolated gate driver circuit (e.g., the gate driver circuit **160** of FIG. **1**) to drive the first MOSFET **415** and the second MOSFET **416**.

(36) According to various embodiments, the driver circuit **419** may provide a driving signal for controlling the first MOSFET **415** and the second MOSFET **416** to the first MOSFET **415** and the second MOSFET **416**. When the magnitude of the driving signal is sufficiently large such that gate voltages of the first MOSFET **415** and the second MOSFET **416** exceed threshold voltages of the first MOSFET **415** and the second MOSFET **416**, the first MOSFET **415** and the second MOSFET **416** may be in the “on” state. In this case, it may be expressed that the driver circuit **419** switches the first MOSFET **415** and the second MOSFET **416** to be in the “on” state. When the magnitude of the driving signal is not sufficiently large such that the gate voltages of the first MOSFET **415** and the second MOSFET **416** are less than or equal to the threshold voltages of the first MOSFET **415** and the second MOSFET **416**, the first MOSFET **415** and the second MOSFET **416** may be in the “off” state. In this case, it may be expressed that the driver circuit **419** switches the first MOSFET **415** and the second MOSFET **416** to be in the “off” state.

(37) FIG. **5A** is a diagram illustrating an equivalent circuit of the wireless power receiver **401** when the driver circuit **419** switches the first MOSFET **415** and the second MOSFET **416** to be in the “on” state. In FIG. **5A**, a receiving circuit **510** may refer to components other than a resonance circuit among the components of the wireless power receiver **401**. When the driver circuit **419** switches the first MOSFET **415** and the second MOSFET **416** to be in the “on” state, the wireless power receiver **401** may include a first circuit **501a** forming a closed loop and a second circuit **502a** forming a closed loop. Based on a magnetic field generated from a wireless power transmitter, a current may be induced in the first circuit **501a** including the first coil **411** and the first capacitor **412**. Based on a magnetic field generated in the first coil **411**, a current may be induced in the second circuit **502a** including the second coil **413**, the third coil **414**, the second capacitor **417**, and the third capacitor **418**. Voltages induced in the second coil **413** and the third coil **414** may be a voltage induced in the first coil **411** that is increased by the goodness Q of the resonance circuit. As a result, the voltage applied to the receiving circuit **510** in FIG. **5A** may be higher than the voltage in FIG. **5B**.

(38) FIG. **5B** is a diagram illustrating an equivalent circuit of the wireless power receiver **401** when the driver circuit **419** switches the first MOSFET **415** and the second MOSFET **416** to be in the “off” state. In FIG. **5B**, the receiving circuit **510** may refer to components other than a resonance circuit among the components of the wireless power receiver **401**. Referring to FIG. **5B**, when the driver circuit **419** switches the first MOSFET **415** and the second MOSFET **416** to be in the “off” state, the wireless power receiver **401** is equivalent to a third circuit **501b** forming one large, closed loop. Based on a magnetic field generated from the wireless power transmitter, a current may be induced in the third circuit **501b** including the first coil **411**, the first capacitor **412**, the second coil **413**, the third coil **414**, the second capacitor **417**, and the third capacitor **418**.

(39) As described above, as the driver circuit **419** switches between on/off in the first MOSFET **415** and the second MOSFET **416**, the voltage applied to the receiving circuit **510** may be different. The driver circuit **419** may control on/off of the first MOSFET **415** and the second MOSFET **416**, based on the voltage applied to the receiving circuit **510**. For example, the driver circuit **419** may switch the first MOSFET **415** and the second MOSFET **416** to the “off” state when the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **420** is greater than or equal to a threshold, and may switch the first MOSFET **415** and the second MOSFET **416** to the “on” state when the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **420** is less than the threshold.

(40) Referring back to FIG. **4**, the rectifier circuit **420** may include four diodes **421**, **422**, **423**, and **424** forming a full bridge structure. According to various embodiments, unlike in FIG. **4**, the rectifier circuit **420** may include four MOSFETs forming a full bridge structure. When the rectifier

circuit **420** includes four MOSFETs, the on/off state of each of the four MOSFETs may be synchronously switched according to the voltage applied to each of the four MOSFETs.

(41) According to various embodiments, unlike in FIG. 4, the rectifier circuit **420** may include two FETs or two diodes forming a half bridge structure.

(42) FIG. 6 is a circuit diagram illustrating a wireless power receiver, according to an embodiment of the disclosure. FIGS. 7A and 7B are diagrams illustrating equivalent circuits of a wireless power receiver, according to an embodiment of the disclosure.

(43) Referring to FIG. 6, a wireless power receiver **601** may include a resonance circuit **610**, a rectifier circuit **630**, a capacitor **640**, and a load **650**.

(44) According to various embodiments, the resonance circuit may include a first coil **611**, a first capacitor **612**, a second coil **618**, a third coil **619**, a second capacitor **621**, a third capacitor **622**, a first MOSFET **615**, a second MOSFET **616**, a third MOSFET **613**, a fourth MOSFET **614**, and a driver circuit **617**. The second coil **618** and the third coil **619** may be wound such that a dot direction is indicated opposite to a direction shown in FIG. 6.

(45) According to various embodiments, unlike in FIG. 6, the first MOSFET **615** and the second MOSFET **616** may be replaced with a first diode and a second diode.

(46) According to various embodiments, the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, the fourth MOSFET **614** may form a full bridge structure. Drains of the first MOSFET **615** and the second MOSFET **616** may be connected to an output end of the rectifier circuit **630**. Sources of the first MOSFET **613** and the second MOSFET **614** may be connected to the ground. Thus, the wireless power receiver **601** may not need an isolated gate driver circuit (e.g., the gate driver circuit **160** of in FIG. 1) to drive the third MOSFET **613** and the fourth MOSFET **614**. The driver circuit **617** may provide a driving signal for synchronously switching the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** to the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**, respectively. The driver circuit **617** may provide a driving signal for switching the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** to the “off” state to the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**, respectively.

(47) FIG. 7A is a diagram illustrating an equivalent circuit of the wireless power receiver **601** when the driver circuit **617** synchronously switches the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** or the driver circuit **617** switches the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** to the “off” state. In FIG. 7A, a component through which a relatively small current flows is indicated by a dotted line.

(48) Referring to FIG. 7A, when the driver circuit **617** synchronously switches the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**, a current may be induced in a first circuit **710a** including the first coil **611** and the first capacitor **612**, based on the magnetic field generated from the wireless power transmitter. The current induced in the first circuit **710a** is AC, such that according to the sign of the induced current, the driver circuit may switch on the first MOSFET **615** and the fourth MOSFET **614** and may switch off the second MOSFET **616** and the third MOSFET **613**, or may switch on the second MOSFET **616** and the third MOSFET **613** and may switch off the first MOSFET **615** and the fourth MOSFET **614**. When the driver circuit **617** synchronously switches the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**, a current smaller than the current flowing in the first circuit **710a** may flow through the second coil **618**, the third coil **619**, the second capacitor **621**, the third capacitor **622**, and the rectifier circuit **630**. The magnitude of the current flowing in the second coil **618** may be smaller as the magnitude of a mutual inductance between the first coil **611** and the second coil **618** decreases. As resonant frequencies of the second coil **618** and the second capacitor **621** are much lower than an operating frequency of the wireless power receiver

601, the magnitude of the current flowing in the second coil **618** may be smaller. The magnitude of the current flowing in the third coil **619** may be smaller as the magnitude of a mutual inductance between the first coil **611** and the third coil **619** decreases. As resonant frequencies of the third coil **619** and the third capacitor **622** are much lower than the operating frequency of the wireless power receiver **601**, the magnitude of the current flowing in the third coil **619** may be smaller.

(49) Referring to FIG. 7A, the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** may form the full bridge structure and drains of the first MOSFET **615** and the second MOSFET **616** may be connected to the output end of the rectifier circuit **630**, such that power induced in the first circuit **710a** may be rectified by the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**. The rectified power may be supplied to the output end of the rectifier circuit **630** and supplied to the capacitor **640** and the load **650**.

(50) According to various embodiments, when the driver circuit **617** switches the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** to the “off” state, power induced in the first circuit **710a** may be rectified by body diodes of the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**. The rectified power may be supplied to the output end of the rectifier circuit **630** and supplied to the capacitor **640** and the load **650**.

(51) According to various embodiments, the driver circuit **617** may switch the third MOSFET **613** and the fourth MOSFET **614** to the “on” state and switch the first MOSFET **615** and the second MOSFET **616** to the “off” state. FIG. 7B is a diagram illustrating an equivalent circuit of the wireless power receiver **601** when the driver circuit **617** switches the third MOSFET **613** and the fourth MOSFET **614** to the “on” state and switches the first MOSFET **615** and the second MOSFET **616** to the “off” state. In FIG. 7B, a component through which a current does not flow is indicated by a dotted line.

(52) Referring to FIG. 7B, as a current flows through the third MOSFET **613** and the fourth MOSFET **614** switched to the “on” state, the resonance circuit may include a second circuit **710b**, forming a closed loop of the first coil **611**, the first capacitor **612**, the third MOSFET **613**, and the fourth MOSFET **614**, and a third circuit **720b**, including the third MOSFET **613**, the fourth MOSFET **614**, the second coil **618**, the third coil **619**, the second capacitor **621**, and the third capacitor **622**. That is, a current may be induced in the second circuit **710b** based on the magnetic field generated from the wireless power transmitter and a current may be induced in the third circuit **720b** based on the magnetic field generated by the first coil **611**, such that the resonance circuit may receive wireless power from the wireless power transmitter. The received wireless power may be rectified by the rectifier circuit **630**, and the rectified wireless power may be supplied to the capacitor **640** and the load **650**.

(53) As described above, as the driver circuit **617** switches between on/off for each of the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**, an operation of the resonance circuit may change.

(54) According to various embodiments, the driver circuit **617** may switch between on/off for each of the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**, based on the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **630**. For example, the driver circuit **617** may synchronously switch the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** when the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **630** is greater than or equal to a threshold. The driver circuit **617** may switch the third MOSFET **613** and the fourth MOSFET **614** to the “on” state and switch the first MOSFET **615** and the second MOSFET **616** to the “off” state, when the voltage $V_{sub.rec}$ of the output end of the rectifier circuit **630** is less than the threshold.

(55) Referring back to FIG. 6, the rectifier circuit **630** may include four diodes **631**, **632**, **633**, and **634** forming the full bridge structure. According to various embodiments, unlike in FIG. 6, the

rectifier circuit **630** may include two FETs or two diodes forming the half bridge structure.

(56) According to various embodiments, unlike in FIG. **6**, the rectifier circuit **630** may include four MOSFETs forming a full bridge structure. When the rectifier circuit **630** includes four MOSFETs, the on/off state of each of the four MOSFETs may be synchronously switched according to the voltage applied to each of the four MOSFETs. For example, four MOSFETs included in the rectifier circuit **630** may be switched by the driver circuit **617**. When the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **630** is greater than or equal to the threshold, the driver circuit **617** may switch four MOSFETs included in the rectifier circuit **630** to the “off” state while synchronously switching the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**. Alternatively, when the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **630** is less than the threshold, the driver circuit **617** may switch the third MOSFET **613** and the fourth MOSFET **614** to the “on” state, switch the first MOSFET **615** and the second MOSFET **616** to the “off” state, and synchronously switch the four MOSFETs included in the rectifier circuit **630**.

(57) According to various embodiments, the load **650** may refer to components other than the resonance circuit **610**, the rectifier circuit **630**, and the capacitor **640** among the components of the wireless power receiver **601**. For example, the load **650** may include a DC/DC converter, an LDO regulator, a charger, and a battery.

(58) FIG. **8** is a diagram illustrating components for controlling a driver circuit of a wireless power receiver, according to an embodiment of the disclosure. A driver circuit **850** of a wireless power receiver (e.g., the wireless power receiver **601** of FIG. **6**) may output driving signals **851**, **852**, **853**, and **854** for controlling on/off of a plurality of switches (e.g., the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** of FIG. **6**) based on a control signal from a micro controller unit (MCU) **830**. The control signal may indicate whether to synchronously switch the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**, whether to switch the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** to the “off” state, or whether to switch the third MOSFET **613** and the fourth MOSFET **614** to the “on” state and switch the first MOSFET **615** and the second MOSFET **616** to the “off” state.

(59) According to various embodiments, the MCU **830** may provide a control signal to the driver circuit **850** on the basis of an analog signal based on the voltage $V_{sub.rec}$ output from a rectifier circuit (e.g., the rectifier circuit **630** of FIG. **6**). The analog signal input to the MCU **830** may be the output voltage $V_{sub.rec}$ of the rectifier circuit passing through a voltage divider **810** including a first resistor **811** and a second resistor **812**, a filter **821**, and a buffer **822**.

(60) According to various embodiments, a hysteresis comparator **870** may provide an interrupt signal based on the output voltage $V_{sub.rec}$ of the rectifier circuit (e.g., the rectifier circuit **630** of FIG. **6**) to at least one of the MCU **830** or the driver circuit **850**.

(61) According to various embodiments, the MCU **830** may provide a control signal to the driver circuit **850** by using an interrupt signal in addition to the analog signal from the buffer **822**.

(62) According to various embodiments, the driver circuit **850** may be provided with the interrupt signal based on the output voltage $V_{sub.rec}$ of the rectifier circuit (e.g., the rectifier circuit **630** of FIG. **6**) from the hysteresis comparator **870**, and output the driving signals **851**, **852**, **853**, and **854** based on the interrupt signal and the control signal.

(63) According to various embodiments, the interrupt signal may be a voltage output from the hysteresis comparator **870** when the output voltage $V_{sub.rec}$ of the rectifier circuit having passed through the voltage divider **860** including the third resistor **861** and the fourth resistor **862** is input to the hysteresis comparator **870**. The hysteresis comparator **870** may output a first interrupt signal when the input voltage is lower than $V_{sub.high}$ and then higher than $V_{sub.high}$, and output a second interrupt signal when the input voltage is higher than $V_{sub.low}$ and then lower than $V_{sub.low}$. $V_{sub.high}$ may be set higher than $V_{sub.low}$.

(64) According to various embodiments, when the driver circuit **850** receives the first interrupt signal from the hysteresis comparator **870**, regardless of a control signal from the MCU **830**, the driver circuit **850** may synchronously switch the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**, or switch the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614** to the “off” state. When the driver circuit **850** receives the second interrupt signal from the hysteresis comparator **870**, regardless of the control signal from the MCU **830**, the driver circuit **850** may switch the third MOSFET **613** and the fourth MOSFET **614** to the “on” state and switch the first MOSFET **615** and the second MOSFET **616** to the “off” state.

(65) FIG. **8** illustrates the voltage divider **810**, the filter **821**, the buffer **822**, the voltage divider **860**, and the hysteresis comparator **870**, but the wireless power receiver (e.g., the wireless power receiver **601** of FIG. **6**) may not include the voltage divider **810**, the filter **821**, and the buffer **822**, or may not include the voltage divider **860** and the hysteresis comparator **870**.

(66) Although the above description has been mainly described with reference to the components of FIG. **6**, the components driving the driver circuit **850** of FIG. **8** may be similarly included in the wireless power receiver **302** of FIG. **3**, the wireless power receiver **401** of FIG. **4**, a wireless power receiver **901** of FIG. **9**, or a wireless power receiver **1101** of FIG. **11**. For example, when the components of FIG. **8** are included in the wireless power receiver **401** of FIG. **4**, the control signal may indicate whether to switch the first MOSFET **415** and the second MOSFET **416** to be in the “on” state or whether to switch the first MOSFET **415** and the second MOSFET **416** to be in the “off” state. In addition, the number of driving signals output from the driver circuit **850** may not be limited to 4.

(67) FIG. **9** is a circuit diagram illustrating a wireless power receiver, according to an embodiment of the disclosure. FIGS. **10A** and **10B** are diagrams illustrating equivalent circuits of the wireless power receiver shown in FIG. **9**, according to an embodiment of the disclosure.

(68) Referring to FIG. **9**, a wireless power receiver **901** may include a resonance circuit **910**, a rectifier circuit **920**, a capacitor **930**, and a load **940**.

(69) According to various embodiments, the resonance circuit may include a first coil **911**, a first capacitor **912**, a second coil **918**, a second capacitor **919**, a first MOSFET **915**, a second MOSFET **916**, a third MOSFET **913**, a fourth MOSFET **914**, and a driver circuit **917**. The second coil **918** may be wound such that a dot direction is indicated opposite to a direction shown in FIG. **9**.

(70) According to various embodiments, the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, the fourth MOSFET **914** may form the full bridge structure. Drains of the first MOSFET **915** and the second MOSFET **916** may be connected to an output end of the rectifier circuit **920**. Sources of the first MOSFET **913** and the second MOSFET **914** may be connected to the ground. Thus, the wireless power receiver **901** may not need an isolated gate driver circuit (e.g., the gate driver circuit **160** of FIG. **1**) to drive the third MOSFET **913** and the fourth MOSFET **914**.

(71) According to various embodiments, the driver circuit **917** may provide a driving signal for synchronously switching the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914** to the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914**, respectively. The driver circuit **917** may provide a driving signal for switching the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914** to the “off” state to the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914**, respectively. FIG. **10A** shows an equivalent circuit of the wireless power receiver **901** when the driver circuit **917** synchronously switches the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914**, or switches the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914** to the “off” state. In FIG. **10A**, a component through which a relatively small current flows is indicated by a dotted line.

(72) Referring to FIG. **10A**, when the driver circuit **917** synchronously switches the first MOSFET

915, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914**, a current may be induced in a first circuit **1010a** including the first coil **911** and the first capacitor **912**, based on the magnetic field generated from the wireless power transmitter. The current induced in the first circuit **1010a** is AC, such that according to the sign of the induced current, the driver circuit may switch on the first MOSFET **915** and the fourth MOSFET **914** and switch off the second MOSFET **916** and the third MOSFET **913**, or may switch on the second MOSFET **916** and the third MOSFET **913** and switch off the first MOSFET **915** and the fourth MOSFET **914**.

(73) Referring to FIG. **10A**, when the driver circuit **917** synchronously switches the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914**, a current smaller than a current flowing in the first circuit **1010a** may flow through the second coil **918**, the second capacitor **919**, and the rectifier circuit **920**. The magnitude of the current flowing in the second coil **918** may decrease as the magnitude of a mutual inductance between the first coil **911** and the second coil **918** decreases. As resonant frequencies of the second coil **918** and the second capacitor **919** are much lower than an operating frequency of the wireless power receiver **901**, the magnitude of the current flowing in the second coil **918** may be smaller.

(74) Referring to FIG. **10A**, the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914** may form the full bridge structure, and drains of the first MOSFET **915** and the second MOSFET **916** may be connected to the output end of the rectifier circuit **920**, such that power induced in the first circuit **710a** may be rectified by the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914**. The rectified power may be supplied to the output end of the rectifier circuit **920** and supplied to the capacitor **930** and the load **940**.

(75) According to various embodiments, when the driver circuit **917** switches the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914** to the “off” state, power induced in the first circuit **710a** may be rectified by body diodes of the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914**. The rectified power may be supplied to the output end of the rectifier circuit **920** and supplied to the capacitor **930** and the load **940**.

(76) According to various embodiments, the driver circuit **917** may switch the third MOSFET **913** and the fourth MOSFET **914** to the “on” state, and switch the first MOSFET **915** and the second MOSFET **916** to the “off” state. FIG. **10B** is a diagram illustrating an equivalent circuit of the wireless power receiver **901** when the driver circuit **917** switches the third MOSFET **913** and the fourth MOSFET **914** to the “on” state and switches the first MOSFET **915** and the second MOSFET **916** to the “off” state. In FIG. **10B**, a component through which current flows is indicated in black and a component through which current does not flow is indicated by a dotted line.

(77) Referring to FIG. **10B**, as current flows through the third MOSFET **913** and the fourth MOSFET **914** switched to the “on” state, the resonance circuit may include a second circuit **1010b**, forming a closed loop of the first coil **911**, the first capacitor **912**, the third MOSFET **913**, and the fourth MOSFET **914**, and a third circuit **1020b**, including the third MOSFET **913**, the fourth MOSFET **914**, the second coil **918**, and the second capacitor **919**. That is, current may be induced in the second circuit **1010b** based on the magnetic field generated from the wireless power transmitter and current may be induced in the third circuit **1020b** based on the magnetic field generated by the first coil **911**, such that the resonance circuit may receive wireless power from the wireless power transmitter. The received wireless power may be rectified by the rectifier circuit **920**, and the rectified wireless power may be supplied to the capacitor **930** and the load **940**.

(78) As described above, as the driver circuit **917** switches between on/off in each of the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914**, an operation of the resonance circuit may change. The driver circuit **917** may switch between on/off in each of the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth

MOSFET **914**, based on the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **920**. For example, the driver circuit **917** may synchronously switch the first MOSFET **915**, the second MOSFET **916**, the third MOSFET **913**, and the fourth MOSFET **914** when the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **920** is greater than or equal to a threshold. The driver circuit **917** may switch the third MOSFET **913** and the fourth MOSFET **914** to the “on” state and switch the first MOSFET **915** and the second MOSFET **916** to the “off” state, when the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **920** is less than the threshold.

(79) Referring back to FIG. **9**, the rectifier circuit **920** may include two diodes **921** and **922** forming the half bridge structure. According to various embodiments, unlike in FIG. **9**, the rectifier circuit **920** may include two MOSFETs forming the half bridge structure. A detailed description is provided below with respect to FIG. **11** in which the rectifier circuit **920** includes two MOSFETs. According to various embodiments, unlike in FIG. **9**, the rectifier circuit **920** may include four FETs or four diodes forming the full bridge structure.

(80) According to various embodiments, the load **940** may refer to components other than the resonance circuit **910**, the rectifier circuit **920**, and the capacitor **930** among the components of the wireless power receiver **901**. For example, the load **940** may include a DC/DC converter, an LDO regulator, a charger, and a battery.

(81) FIG. **11** is a circuit diagram illustrating a wireless power receiver, according to an embodiment of the disclosure. A wireless power receiver **1101** of FIG. **11** is similar to the wireless power receiver **901** of FIG. **9**, except that the wireless power receiver **1101** further includes a matching network **1150** and a rectifier circuit **1120**, which includes two MOSFETs **1121** and **1122** instead of two diodes. Thus, the foregoing details described regarding the resonance circuit **910**, the driver circuit **917**, the rectifier circuit **920**, the capacitor **930**, and the load **940** of FIG. **9** may be equally applied to a resonance circuit **1110**, a driver circuit **1117**, a rectifier circuit **1120**, a capacitor **1130**, and load **1140** of FIG. **11**.

(82) Referring to FIG. **11**, according to various embodiments, the wireless power receiver **1101** may include a matching network **1150**. The rectifier circuit **1120** has a smaller impedance when having the half bridge structure than when having the full bridge structure, such that the matching network **1150** may be used to compensate for a reduced impedance. The matching network **1150** may include at least one capacitor and at least one inductor. The matching network **1150** may include two capacitors and one inductor, a first end of the matching network **1150** may be a first end of a first capacitor, a second end of the first capacitor may be connected to a first end of a second capacitor and a first end of an inductor, and a second end of the second capacitor and a second end of the inductor may be a second end of the matching network **1150**.

(83) According to various embodiments, the rectifier circuit **1120** may include two MOSFETs **1121** and **1122** forming the half bridge structure. The on/off state of each of the two MOSFETs may be controlled in an alternating manner based on a voltage applied to each of the two MOSFETs.

(84) For example, the two MOSFETs included in the rectifier circuit **920** may be switched by the driver circuit **1117**. When the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **1120** is greater than or equal to the threshold, the driver circuit **1117** may switch the two MOSFETs included in the rectifier circuit **1120** to the “off” state while synchronously switching the first MOSFET **1115**, the second MOSFET **1116**, the third MOSFET **1113**, and the fourth MOSFET **1114** or switching the first MOSFET **1115**, the second MOSFET **1116**, the third MOSFET **1113**, and the fourth MOSFET **1114** to the “off” state. Alternatively, when the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **1120** is less than the threshold, the driver circuit **1117** may switch the third MOSFET **1113** and the fourth MOSFET **1114** to the “on” state, switch the first MOSFET **1115** and the second MOSFET **1116** to the “off” state, and alternately switch the two MOSFETs included in the rectifier circuit **1120**.

(85) Unlike in FIG. **11**, according to various embodiments, the two MOSFETs included in the rectifier circuit **920** may be controlled by a separate driver circuit other than the driver circuit **1117**.

(86) FIG. 12 is a circuit diagram illustrating a wireless power receiver, according to an embodiment of the disclosure. FIGS. 13A and 13B are diagrams illustrating equivalent circuits of the wireless power receiver shown in FIG. 12, according to an embodiment of the disclosure.

(87) Referring to FIG. 12, a wireless power receiver 1201 may include a resonance circuit 1210, a rectifier circuit 1230, a capacitor 1240, and a load 1250.

(88) According to various embodiments, the resonance circuit 1210 may include a first coil 1211, a first capacitor 1212, a second coil 1218, a second capacitor 1219, a first MOSFET 1215, a second MOSFET 1216, a third MOSFET 1213, a fourth MOSFET 1214, and a driver circuit 1217. The second coil 1218 may be wound such that a dot direction is indicated opposite to a direction shown in FIG. 12.

(89) According to various embodiments, the second coil 1218 and the second capacitor 1219 may be connected in series. One end of the second coil 1218 and one end of the second capacitor 1219 that are serially connected may be connected to the ground, and the other end of the second coil 1218 and the other end of the second capacitor 1219 that are serially connected may be connected to the rectifier circuit 1230.

(90) According to various embodiments, the first MOSFET 1215, the second MOSFET 1216, the third MOSFET 1213, and the fourth MOSFET 1214 may form the full bridge structure. Drains of the first MOSFET 1215 and the second MOSFET 1216 may be connected to an output end of the rectifier circuit 1230. Sources of the first MOSFET 1213 and the second MOSFET 1214 may be connected to the ground. Thus, the wireless power receiver 1201 may not need an isolated gate driver circuit (e.g., the gate driver circuit 160 of FIG. 1) to drive the third MOSFET 1213 and the fourth MOSFET 1214.

(91) According to various embodiments, the driver circuit 1217 may provide a driving signal for synchronously switching the first MOSFET 1215, the second MOSFET 1216, the third MOSFET 1213, and the fourth MOSFET 1214 to the first MOSFET 1215, the second MOSFET 1216, the third MOSFET 1213, and the fourth MOSFET 1214, respectively. The driver circuit 1217 may provide a driving signal for switching the first MOSFET 1215, the second MOSFET 1216, the third MOSFET 1213, and the fourth MOSFET 1214 to the “off” state to the first MOSFET 1215, the second MOSFET 1216, the third MOSFET 1213, and the fourth MOSFET 1214, respectively.

(92) FIG. 13A is a diagram illustrating an equivalent circuit of the wireless power receiver 1201 when the driver circuit 1217 synchronously switches the first MOSFET 1215, the second MOSFET 1216, the third MOSFET 1213, and the fourth MOSFET 1214, or the driver circuit 617 switches the first MOSFET 1215, the second MOSFET 1216, the third MOSFET 1213, and the fourth MOSFET 1214 to the “off” state. In FIG. 13A, a component through which a relatively small current flows is indicated by a dotted line.

(93) Referring to FIG. 13A, when the driver circuit 1217 synchronously switches the first MOSFET 1215, the second MOSFET 1216, the third MOSFET 1213, and the fourth MOSFET 1214, current may be induced in a first circuit 1310a including the first coil 1211 and the first capacitor 1212, based on the magnetic field generated from the wireless power transmitter. The current induced in the first circuit 1310a is AC, such that according to the sign of the induced current, the driver circuit may switch on the first MOSFET 1215 and the fourth MOSFET 1214 and switch off the second MOSFET 1216 and the third MOSFET 1213, or may switch on the second MOSFET 1216 and the third MOSFET 1213 and switch off the first MOSFET 1215 and the fourth MOSFET 1214.

(94) Referring to FIG. 13A, when the driver circuit 1217 synchronously switches the first MOSFET 1215, the second MOSFET 1216, the third MOSFET 1213, and the fourth MOSFET 1214, a current smaller than a current flowing in the first circuit 1310a may flow through the second coil 1218, the second capacitor 1219, and the rectifier circuit 1230. The magnitude of the current flowing in the second coil 1218 may decrease as the magnitude of a mutual inductance $M_{sub.2f}$ between the first coil 1211 and the second coil 1218 decreases. As resonant frequencies of

the second coil **1218** and the second capacitor **1219** are much lower than an operating frequency of the wireless power receiver **1201**, the magnitude of the current flowing in the second coil **1218** may be smaller.

(95) Referring to FIG. **13A**, the first MOSFET **1215**, the second MOSFET **1216**, the third MOSFET **1213**, and the fourth MOSFET **1214** may form the full bridge structure and drains of the first MOSFET **1215** and the second MOSFET **1216** are connected to the output end of the rectifier circuit **1230**, such that power induced in the first circuit **1310a** may be rectified by the first MOSFET **1215**, the second MOSFET **1216**, the third MOSFET **1213**, and the fourth MOSFET **1214**. The rectified power may be supplied to the output end of the rectifier circuit **1230** and supplied to the capacitor **1240** and the load **1250**.

(96) According to various embodiments, when the driver circuit **1217** switches the first MOSFET **1215**, the second MOSFET **1216**, the third MOSFET **1213**, and the fourth MOSFET **1214** to the “off” state, power induced in the first circuit **710a** may be rectified by body diodes of the first MOSFET **1215**, the second MOSFET **1216**, the third MOSFET **1213**, and the fourth MOSFET **1214**. The rectified power may be supplied to the output end of the rectifier circuit **1230** and supplied to the capacitor **1240** and the load **1250**.

(97) According to various embodiments, the driver circuit **1217** may switch the third MOSFET **1213** and the fourth MOSFET **1214** to the “on” state and switch the first MOSFET **1215** and the second MOSFET **1216** to the “off” state.

(98) FIG. **13B** is a diagram illustrating an equivalent circuit of the wireless power receiver **1201** when the driver circuit **1217** switches the third MOSFET **1213** and the fourth MOSFET **1214** to the “on” state and switches the first MOSFET **1215** and the second MOSFET **1216** to the “off” state. In FIG. **13B**, a component through which a current flows is indicated in black and a component through which a current does not flow is indicated by a dotted line.

(99) Referring to FIG. **13B**, as a current flows through the third MOSFET **1213** and the fourth MOSFET **1214** switched to the “on” state, the resonance circuit may include a second circuit **1310b**, forming a closed loop of the first coil **1211**, the first capacitor **1212**, the third MOSFET **1213**, and the fourth MOSFET **1214**, and a third circuit **1320b** including the second coil **1218** and the second capacitor **1219**. That is, a current may be induced in the second circuit **1310b** based on the magnetic field generated from the wireless power transmitter and a current may be induced in the third circuit **1320b** based on the magnetic field generated by the first coil **1211**, such that the resonance circuit may receive wireless power from the wireless power transmitter. The received wireless power may be rectified by the rectifier circuit **1230**, and the rectified wireless power may be supplied to the capacitor **1240** and the load **1250**.

(100) As described above, as the driver circuit **1217** switches between on/off in each of the first MOSFET **1215**, the second MOSFET **1216**, the third MOSFET **1213**, and the fourth MOSFET **1214**, an operation of the resonance circuit may change. The driver circuit **1217** may control on/off in each of the first MOSFET **1213**, the second MOSFET **1216**, the third MOSFET **613**, and the fourth MOSFET **1214**, based on the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **1230**. For example, the driver circuit **1217** may synchronously switch the first MOSFET **1215**, the second MOSFET **1216**, the third MOSFET **1213**, and the fourth MOSFET **1214** when the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **1230** is greater than or equal to a threshold. The driver circuit **1217** may switch the third MOSFET **1213** and the fourth MOSFET **1214** to the “on” state and switch the first MOSFET **1215** and the second MOSFET **1216** to the “off” state, when the voltage $V_{sub.rec}$ at the output end of the rectifier circuit **1230** is less than the threshold.

(101) Referring back to FIG. **12**, the rectifier circuit **1230** may include two diodes **1231** and **1232** forming the half bridge structure. According to various embodiments, unlike in FIG. **12**, the rectifier circuit **1230** may include two MOSFETs forming the half bridge structure. According to various embodiments, unlike in FIG. **12**, the rectifier circuit **1230** may include four FETs or four diodes forming the full bridge structure.

(102) According to various embodiments, the load **1250** may refer to any component other than the resonance circuit **1210**, the rectifier circuit **1230**, and the capacitor **1240** among the components of the wireless power receiver **1201**. For example, the load **1250** may include a DC/DC converter, an LDO regulator, a charger, and a battery.

(103) According to various embodiments, the wireless power receiver **601** may receive wireless power from a wireless power transmitter. The wireless power receiver **601** may include a resonance circuit configured to receive the wireless power, and including the first coil **611**, the second coil **618**, and the first capacitor **612**. The wireless power receiver **601** may include a rectifier circuit connected to the resonance circuit, and including a first rectifier circuit and a second rectifier circuit **620** forming a full bridge structure. The wireless power receiver **601** may include the driver circuit **617**. The first rectifier circuit may include the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**. Sources of the first MOSFET **615** and the second MOSFET **616** are connected to both ends of a resonator in which the first coil **611** and the first capacitor **612** are connected in series. Sources of the third MOSFET **613** and the fourth MOSFET **614** are connected to ground. The driver circuit **617** is connected to gates of the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**. When the driver circuit **617** switches off the first MOSFET **615** and the second MOSFET **616** and switches on the third MOSFET **613** and the fourth MOSFET **614**, as a first current is induced in the resonator based on a first magnetic field generated from the wireless power transmitter and a second current is induced in the second coil **618** based on a second magnetic field generated in the resonator, the resonance circuit receives the wireless power, and the current induced in the second coil **618** may be rectified by the second rectifier circuit **630**.

(104) According to various embodiments, the second rectifier circuit **630** may include four FETs or four diodes **631**, **632**, **633**, and **634** forming a full bridge structure.

(105) According to various embodiments, the second rectifier circuit **630** may include two FETs or two diodes **921** and **922** forming a half bridge structure.

(106) According to various embodiments, when the driver circuit synchronously switches the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**, as the first current is induced based on the first magnetic field generated from the wireless power transmitter, the resonator may receive the wireless power, and the received wireless power may be rectified by the first rectifier circuit.

(107) According to various embodiments, when the driver circuit **617** switches off the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, the fourth MOSFET **614**, as the first current is induced based on the first magnetic field generated from the wireless power transmitter, the resonator may receive the wireless power, and the received wireless power may be rectified by body diodes of the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613**, and the fourth MOSFET **614**.

(108) According to various embodiments, when the driver circuit **617** switches off the first MOSFET **615** and the second MOSFET **616** and switches on the third MOSFET **613** and the fourth MOSFET **614**, the driver circuit **617** may be configured to synchronously switch on or off the two FETs included in the second rectifier circuit **630**.

(109) According to various embodiments, the wireless power receiver may further include the matching network **1150** connected between the second coil **617** and the second rectifier circuit **630**, in which the matching network **1150** may include at least one capacitor and at least one inductor.

(110) According to various embodiments, the wireless power receiver may further include the processor **830**, in which the processor **830** may be configured to control the driver circuit **617** based on the rectified wireless power.

(111) According to various embodiments, the wireless power receiver may further include the hysteresis comparator **870**, in which the hysteresis comparator **870** may be configured to generate an interrupt signal based on the rectified wireless power, and the processor **830** may be configured

to control the driver circuit **617** based on the interrupt signal.

(112) According to various embodiments, the processor **830** may be configured to control the driver circuit **617** to synchronously switch the first MOSFET **615**, the second MOSFET **616**, the third MOSFET **613** and the fourth MOSFET **614** or to switch off the first MOSFET **615**, the second MOSFET **615**, the third MOSFET **613**, and the fourth MOSFET **614**, when a voltage of the rectified wireless power is greater than or equal to a threshold.

(113) According to various embodiments, the processor **830** may be configured to control the driver circuit **617** to switch off the first MOSFET **615** and the second MOSFET **616** and to switch on the third MOSFET **613** and the fourth MOSFET **614**, when the voltage of the rectified wireless power is less than the threshold.

(114) According to various embodiments, the resonance circuit may further include the second capacitor **621**, the third coil **619**, and the third capacitor **622**. The first end of the second coil **618** may be connected to the drain of the third MOSFET **613**. The second end of the second coil **618** may be connected to the first end of the second capacitor **621**. The second end of the second capacitor **621** may be connected to the second rectifier circuit **630**. The first end of the third coil **619** may be connected to the drain of the fourth MOSFET **614**. The second end of the third coil **619** is connected to the first end of the third capacitor **622**. The second end of the third capacitor **622** may be connected to the second rectifier circuit **630**.

(115) According to various embodiments, the drains of the first MOSFET **615** and the second MOSFET **616** may be connected to the output end of the rectifier circuit **630**.

(116) According to various embodiments, the resonance circuit may further include the second capacitor **919**. The first end of the second coil **918** may be connected to the drain of the third MOSFET **913**. The second end of the second coil **918** may be connected to the first end of the second capacitor **919**. The second end of the second capacitor **919** may be connected to the second rectifier circuit **920**.

(117) According to various embodiments, the first rectifier circuit may include a first diode and a second diode in place of the first MOSFET **615** and the second MOSFET **616**.

(118) According to various embodiments, the wireless power receiver **401** may receive wireless power from a wireless power transmitter. The wireless power receiver **401** may include a resonance circuit configured to receive the wireless power, and including the first coil **411**, the second coil **413**, the first capacitor **412**, and a bidirectional switch. The wireless power receiver **401** may include the rectifier circuit **420** connected to the resonance circuit, and a driver circuit **419**. The bidirectional switch may include the first MOSFET **415** and the second MOSFET **416**. Sources of the first MOSFET **415** and the second MOSFET **416** may be connected to the ground. The driver circuit **419** may be connected to the gates of the first MOSFET **415** and the second MOSFET **416**. When the driver circuit **419** switches off the first MOSFET **415** and the second MOSFET **416**, as a first current is induced to a first circuit including the first coil **411**, the second coil **413**, and the first capacitor **412** based on a first magnetic field generated from the wireless power transmitter, the resonance circuit may be configured to receive the wireless power. When the driver circuit **419** switches on the first MOSFET **415** and the second MOSFET **416**, as a second current is induced in a second circuit including the first coil **411** and the first capacitor **412** based on the first magnetic field generated from the wireless power transmitter and a third current is induced in a third circuit including the second coil **413** based on a second magnetic field generated in the second circuit, the resonance circuit may be configured to receive the wireless power.

(119) According to various embodiments, the rectifier circuit **420** may include four FETs or four diodes **421**, **422**, **423**, and **424** forming a full bridge structure.

(120) According to various embodiments, the rectifier circuit **420** may include two FETs or two diodes forming a half bridge structure.

(121) According to various embodiments, the wireless power receiver **401** may further include a matching network connected to the rectifier circuit **420**, in which the matching network may

include at least one capacitor and at least one inductor.

(122) According to various embodiments, the wireless power receiver **401** may further include a processor, in which the processor may be configured to control the driver circuit **419** based on the rectified wireless power.

(123) The wireless power receiver according to various embodiments disclosed herein may be one of various types of devices. The wireless power receiver may include, for example, a portable communication device (e.g., a smart phone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. The wireless power receiver according to an embodiment of the disclosure may not be limited to the above-listed devices.

(124) As used in various embodiments, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(125) Various embodiments as set forth herein may be implemented as software (e.g., program including one or more instructions that are stored in a storage medium (e.g., internal memory or external memory) that is readable by a machine (e.g., electronic device). For example, a processor of the machine (e.g., electronic device) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(126) According to an embodiment, a method according to various embodiments disclosed herein may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smart phones) directly. When distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(127) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or a plurality of entities, and some of the plurality of entities may be separately disposed on different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(128) While the disclosure has been shown and described with reference to certain embodiments thereof, it will be understood by those skilled in the art that various changes in form and detail may be made therein without departing from the scope of the disclosure. Therefore, the scope of the disclosure should not be defined as being limited to the embodiments, but should be defined by the appended claims and equivalents thereof.

Claims

1. A wireless power receiver comprising: a resonance circuit comprising a first coil, a first capacitor, a second coil, a second capacitor, a third coil, and a third capacitor; a switching circuit comprising a first switch, and a second switch, wherein the first switch and a second switch are a n-type field-effect transistor (FET) and a first source of the first switch is connected to a second source of the second switch; a rectifier circuit coupled to the resonance circuit and configured to converting alternating current (AC) to direct current (DC) power; and a driver circuit is configured to: control the switching circuit to switch on the first switch and the second switch to form a first resonator in which the first coil and the first capacitor are connected in series, wherein a first current is induced in the first resonator based on a first electromagnetic field from an external wireless power transmitter, and a second current is induced in the second coil and the third coil based on a second electromagnetic field generated by the first current, and control the switching circuit to switch off the first switch and the second switch to form a second resonator in which the first coil, the first capacitor, the second coil, the second capacitor, the third coil, and the third capacitor are connected in series, wherein a third current is induced in the second resonator based on a third electromagnetic field from the external wireless power transmitter.
 2. The wireless power receiver of claim 1, wherein the rectifier circuit comprises four field effect transistors (FETs) or four diodes forming a full bridge structure.
 3. The wireless power receiver of claim 1, wherein the rectifier circuit comprises two FETs or two diodes forming a half bridge structure.
 4. The wireless power receiver of claim 3, further comprising a matching network connected to the rectifier circuit, wherein the matching network includes at least one capacitor and at least one inductor.
 5. The wireless power receiver of claim 1, further comprising a processor, wherein the processor is configured to control the driver circuit based on power rectified by the rectifier circuit.
 6. The wireless power receiver of claim 5, wherein the processor is configured to control the driver circuit to control the switching circuit to switch off the first switch and the second switch, based on a voltage of the rectified power being greater than or equal to a threshold.
 7. The wireless power receiver of claim 5, wherein the processor is configured to control the driver circuit to control the switching circuit to switch on the first switch and the second switch, based on a voltage of the rectified power being less than a threshold.
 8. The wireless power receiver of claim 1, wherein the first source and the second source are connected to ground, and wherein the driver circuit is connected to gates of the first MOSFET and the second MOSFET.
 9. The wireless power receiver of claim 1, wherein: a first end of the second coil is connected to a first drain of the first switch, a second end of the second coil is connected to a first end of the second capacitor, a second end of the second capacitor is connected to the rectifier circuit, a first end of the third coil is connected to a second drain of the second switch, a second end of the third coil is connected to a first end of the third capacitor, and a second end of the third capacitor is connected to the rectifier circuit.
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